The Himalayan Tightrope

Balancing the Economy-Environment Trade-off

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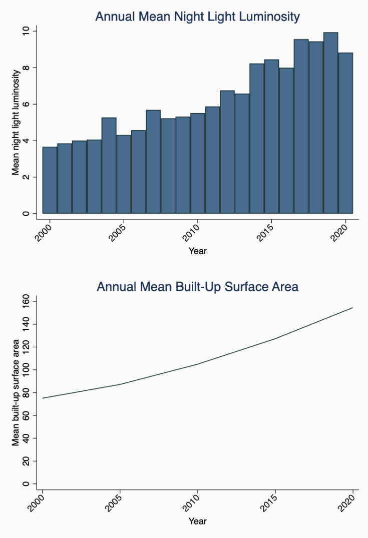
The Trade-off Exists...

- Why build infrastructure?
  - Network effects, economic growth
  - Administrative, defense reasons
  - Political incentives



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  - Network effects, economic growth
  - Administrative, defense reasons
  - Political incentives
- But there is a growing *long-run* and *human* cost.

Landslides biggest killer in Uttarakhand; highest no. of casualties this year

Himachal cloudbursts: Are hydropower projects to blame?

Four of the 6 places where cloudbursts occurred last week were locations of hydro projects

Man-made disaster in Himachal Pradesh

The result: Monsoon showers triggered flash floods in riverside towns and highways in July and August and caused massive loss of life and property.

Himachal: 4 injured as landslide triggered by rain hits bus in Manali, markets flooded in Kullu

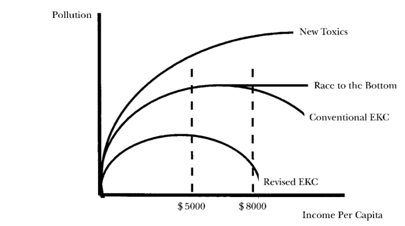
By Shailee Dogra

Frequent Natural Disasters Continue To Haunt Himachal Pradesh, Scientists Call For Urgent 'Risk Avoidance'

...But Do We Measure It Right?

Environmental Kuznets Curve

- Inverted-U relationship between economic growth and environmental degradation.
- Contradictory findings in recent literature.



Different EKC adaptations from Dasgupta et al. (2002).

Questions

- How does *infrastructure-led* growth relate to environmental degradation?
- Does the Environmental Kuznets curve hold (at the subnational level)?
- Do macro-level models hold for:
  - sensitive ecologies?
  - micro-level data?

This Paper

Our answer: Not well.

- We build a high-res geospatial dataset at village/town level
- We estimate the EKC: 5 indicators, 20 years of data
- We focus on two Himalayan states, where incentives exist for both infrastructure investment and ecological sustainability

Data

Dataset Construction

Shrid Our base unit of analysis. SHRUG-DevDataLab. Usually single village/town, can be aggregation if merged/separated between 1990-2013.

Indicators Satellite-based data on infrastructure growth and environmental degradation.

Indices Standardization, then take the:

- arithmetic mean, or
- principal component

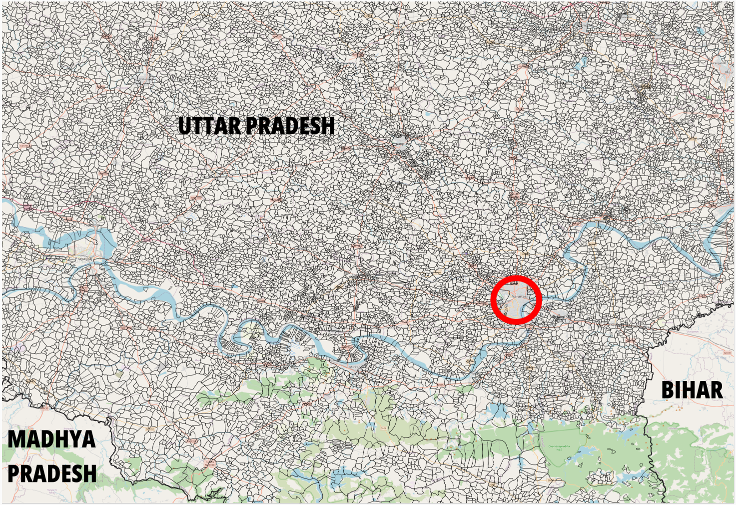


Figure 1: Red circle shows Varanasi.

Economic Indicators

Night lights

Why? Strong correlation with GDP growth even at subnational level (Doll, Muller, and Morley, 2006).

Source NASA, harmonized across satellites by Li, Zhou, and Zhao (2020), yearly data.

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Built-up area

Why? Detects economic activity at highly local levels (Baragwanath et al., 2021).

Source ESA, Copernicus GHSL product, five-year intervals.

Environmental Indicators

PM2.5 air quality

Why? Respected measure of environmental degradation (Caviglia-Harris, Chambers, and Kahn, 2009).

Source SHRUG v2, Development Data Lab cf. Donkelaar et al. (2021).

Environmental Indicators

PM2.5 air quality

Why? Respected measure of environmental degradation (Caviglia-Harris, Chambers, and Kahn, 2009).

Source SHRUG v2, Development Data Lab cf. Donkelaar et al. (2021).

Land surface temperature

Why? Long-run changes from environmental degradation, global warming (European Space Agency, n.d.).

Source NASA MODIS, yearly aggregates from 8-weekly data.

Environmental Indicators

Forest cover

Why? Relevant for infra growth, also used for horizontal devolution (Wang, 2003; XV Finance Commission, 2020).

Source SHRUG v2, Development Data Lab, Terra MODIS (NASA).


Night-time Lights for Himachal Pradesh
VIIRS Black Marble, EarthData.
Generated by author.
LAHUL & SPITI
KINNAUR
KULLU
CHAMBA
SHIMLA
MANDI
KANGRA
HAMIRPUR
BILASPUR
SOLAN
SIRMAUR
UNA

Model and Findings

General Quadratic Model

$$\text{Env}_{it} = \beta_0 + \beta_1(\text{Econ}_{it}) + \beta_1(\text{Econ}^2_{it}) + \eta_i + \gamma_t + \epsilon_{it}$$

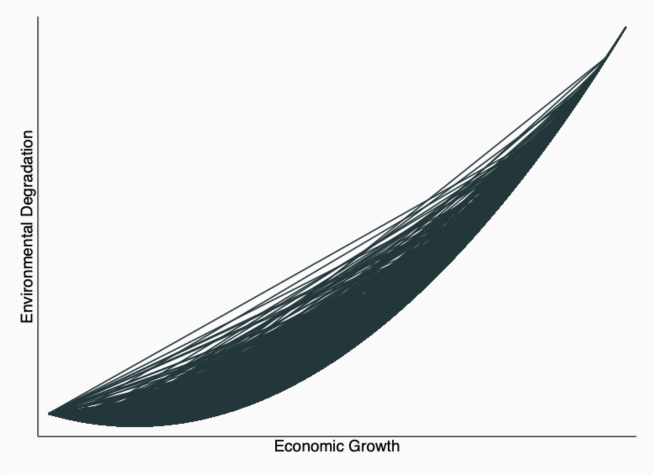
where:

- Env_{it} = environmental index value at area i at time t.
- Econ_{it} = economic index value at area i at time t.
- η_i = *shrid*-level fixed effects.
- γ_i = year fixed effects.
- ϵ_{it} = idiosyncratic error term.

Model 1: Night Lights

	Dependent variable:		
	Environmental Index$_{VCF,PM25}$		
	(1)	(2)	(3)
NightLights	-0.025^{***}	-0.018^{***}	0.270^{***}
	(0.001)	(0.001)	(0.001)
NightLights^2	0.022^{***}	0.016^{***}	0.003^{***}
	(0.0003)	(0.0002)	(0.0003)
PCA	Yes	No	No
TWFE	Yes	Yes	No
Note:			*p<0.1; **p<0.05; ***p<0.01

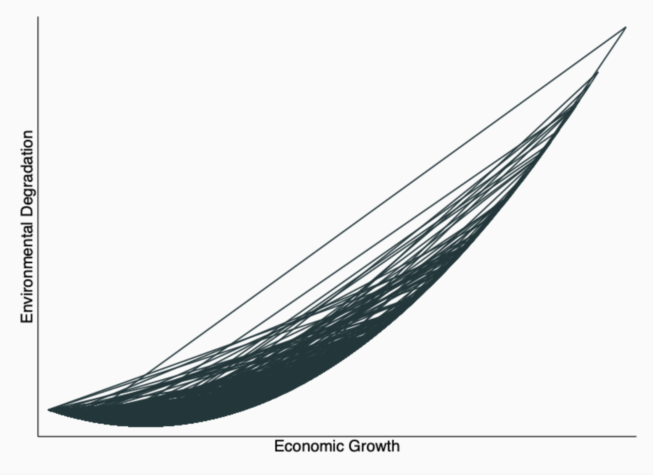
Model 1: Night Lights



Model 2: Adding Built-up Area

	Dependent variable:		
	Environmental Index$_{VCF,PM25}$		
	(1)	(2)	(3)
EconIndex	-0.061^{***}	-0.061^{***}	0.002^{***}
	(0.003)	(0.00004)	(0.00003)
EconIndex^2	0.013^{***}	0.019^{***}	0.003^{***}
	(0.0004)	(0.0003)	(0.0003)
PCA	Yes	No	No
TWFE	Yes	Yes	No
Note:	*p<0.1; **p<0.05; ***p<0.01		

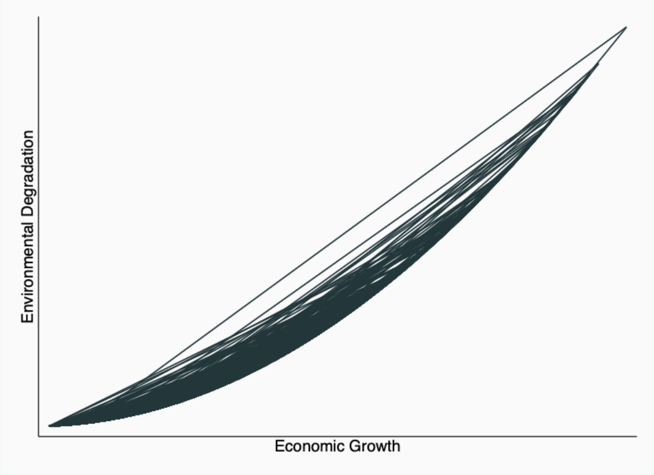
Model 2: Adding Built-up Area



Model 3: Adding Land Surface Temperature

	Dependent variable:		
	Environmental Index_{full}		
	(1)	(2)	(3)
EconIndex	0.038***	−0.061***	0.414***
	(0.003)	(0.003)	(0.003)
EconIndex^2	0.010***	0.019***	−0.022***
	(0.0004)	(0.0005)	(0.001)
PCA	Yes	No	No
TWFE	Yes	Yes	No
Note:			*p<0.1; **p<0.05; ***p<0.01

Model 3: Adding Land Surface Temperature



Bottom Line

- Economy-environment trade-offs in the Himalayas - an ecologically-sensitive region - are complex.
- EKC model fails when applied to localized contexts.
- Future research: pinpoint specific projects to build a treatment group for causal analysis, new estimation methods including GMM, causal machine learning.

Policy Implications

- Centralized policy can have heterogenous outcomes.
- Need for localized, dynamic policy frameworks.
- Underlines the importance of ecological risk assessments.
- Emphasis on renewable energy, sustainable urban planning.

Thank you.

Our dataset is open-sourced at:

github.com/RitwizSarma/himalayan-env

Open to questions/comments.